SINGLE STAGE (12V->48V)
BOOST MODE(PHASE SHEDDING NOT POSSIBLE)

S.NO	Load %	Power demand(W)	R(load) (ohms)	Vin (V)	Iin (A)	Pin (W)	Vout (V)	Iout (A)	Pout (W)	Efficiency (%)
1	100	200	11.52	24	8.74	209.9	47.56	4.129	196.3	93.5
2	80	160	14.40	24	7.071	169.7	47.98	3.332	159.8	94.16
3	60	120	19.20	24	5.56	128.7	48.4	2.521	122	94.79
4	50	100	23.04	24	4.48	107.9	48.62	2.11	102.6	95.08
5	40	80	28.80	24	3.62	87.03	48.84	1.69	82.81	95.15
6	30	60	38.40	24	2.74	65.9	49.06	1.27	62.66	95.08
7	20	40	57.60	24	1.85	44.5	49.23	0.85	42.1	94.6
8	10	20	115.20	24	1.362	32.67	59.18	0.513	30.4	93.05

SINGLE STAGE (48V->12V)
BUCK MODE WITHOUT PHASE SHEDDING

S.NO	Load %	Power demand(W)	R(load) (ohms)	Vin (V)	Iin (A)	Pin (W)	Vout (V)	Iout (A)	Pout (W)	Efficiency (%)
1	100	250	0.576	48	5.45	261.9	11.35	19.7	223.5	85.3
2	80	200	0.72	48	4.4	211.6	11.44	15.89	181.8	85.9
3	60	150	0.96	48	3.34	160.5	11.54	12.02	138.7	86.4
4	50	125	1.152	48	2.8	134.7	11.59	10.06	116.6	86.5
5	40	100	1.44	48	2.26	108.6	11.64	8.082	94.04	86.6
6	30	75	1.92	48	1.71	82.25	11.69	6.087	71.14	86.4
7	20	50	2.88	48	1.16	55.7	11.74	4.071	47.84	85.88
8	10	25	5.76	48	0.6	28.92	11.79	2.04	24.13	83.43

SINGLE STAGE (48V->12V)
BUCK MODE WITH PHASE SHEDDING

S.NO	Load %	Power demand(W)	R(load) (ohms)	Vin (V)	Iin (A)	Pin (W)	Vout (V)	Iout (A)	Pout (W)	Efficiency (%)
1	100	250	0.576	48	5.547	266.3	11.54	20.03	231.2	86.8
2	80	200	0.72	48	4.483	215.3	11.64	16.16	188.1	87.3
3	60	150	0.96	48	3.3413	163.5	11.74	12.22	143.3	87.6
4	50	125	1.152	48	2.77	133	11.53	10	115.3	86.69
5	40	100	1.44	48	2.231	107.1	11.58	8.044	93.17	86.99
6	30	75	1.92	48	1.687	80.99	11.64	6.063	70.58	87.146
7	20	50	2.88	48	1.13	54.58	11.7	4.062	47.53	87.08
8	10	25	5.76	48	0.58	27.91	11.76	2.041	24	85.99

SINGLE STAGE (48V->24V)
BUCK MODE WITHOUT PHASE SHEDDING

S.NO	Load %	Power demand(W)	R(load) (ohms)	Vin (V)	Iin (A)	Pin (W)	Vout (V)	Iout (A)	Pout (W)	Efficiency (%)
1	100	200	2.88	48	4.37	209.9	23.72	8.236	195.4	93.09
2	80	160	3.60	48	3.517	168.9	23.79	6.608	157.2	93.07
3	60	120	4.80	48	2.65	127.6	23.86	4.97	118.6	92.9
4	50	100	5.76	48	2.226	106.9	23.89	4.148	99.08	92.68
5	40	80	7.10	48	1.793	86.11	23.92	3.323	79.5	92.32
6	30	60	9.60	48	1.359	65.27	23.96	2.496	59.79	91.6
7	20	40	14.40	48	0.925	44.46	24	1.667	40.03	90.03
8	10	20	28.80	48	0.67	32.47	29.25	1.016	29.71	90.49

SINGLE STAGE (48V->24V)
BUCK MODE WITH PHASE SHEDDING

S.NO	Load %	Power demand(W)	R(load) (ohms)	Vin (V)	Iin (A)	Pin (W)	Vout (V)	Iout (A)	Pout (W)	Efficiency (%)
1	100	200	2.88	48	4.40	211.3	23.87	8.29	197.9	93.65
2	80	160	3.60	48	3.53	169.8	23.91	6.643	158.9	94.69
3	60	120	4.80	48	2.66	128	23.92	4.982	119.1	93.04
4	50	100	5.76	48	2.193	105.3	23.79	4.131	98.3	93.35
5	40	80	7.10	48	1.762	84.62	23.84	3.311	78.93	93.27
6	30	60	9.60	48	1.33	63.88	23.88	2.48	59.42	93.01
7	20	40	14.40	48	0.89	46.64	23.93	1.662	43.06	92.33
8	10	20	28.80	48	0.46	22.23	24.02	0.8319	20.04	91.8